Tsz Fei Wang

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Education

University of Waterloo | Honours Computer Science (Co-op) — Sept 2026 – May 2031 (expected)

Skills & Technologies

Languages & Frameworks: TypeScript, JavaScript, HTML/CSS, Python, Java, SQL, React, Next.js, Node.js, FastAPI, Three.js
Tools: Git, Docker, GitHub Actions, Firebase, Elasticsearch, Vercel, Cloudflare Workers, Postman, Playwright, Linux

Experience

Krembil Research Institute, World Community Grid — Mar 2026 – Jun 2026
Software Engineer Intern — *Toronto, ON*

- Built most of the frontend for Catrin's full-stack data access portal, including the search, filter and results views researchers use to explore **~4 million** TF-TG associations across **33,000 genes**.
- Reconciled the schemas of **9 independent data sources** so the portal can query them as one dataset.

Century Frontier Asset Management — Jul 2026 – Aug 2026
Quantitative Research Intern — *Shenzhen, CN*

- Designed a point-in-time backtesting pipeline in **Python** with **pandas** and yfinance that only shows each model data from before its prediction date, removing lookahead bias from LLM-driven equity research.
- Built a model-agnostic evaluation framework benchmarking **9 LLM architectures** across **104 tickers**, scoring signal quality separately from stated confidence and showing that confidence was a poor guide to accuracy.

William Lyon Mackenzie C.I. Official Website — Nov 2024 – Present
Web Developer — *Toronto, ON*

- Maintain the official school website used by **1,400+ students**, keeping announcements, pages and navigation current for staff and students.
- Migrated the school's content off a deactivated information portal onto the new site, working with the administration on what to keep and how to organize it.

Mackenzie Cybersecurity Team — Jun 2025 – Jun 2026
Team Captain, President — *Toronto, ON*

- Captained the team to **First in Canada** in CyberTitan, Canada's largest cyberdefense competition, securing Windows, Linux and Cisco systems.

Projects

Physical Studio | *Python, React, Three.js, Docker* | ***Winner**, Hack the Valley 11* — *Oct 2026*

- Engineered the **Python** MCP server that lets an LLM design hardware projects from a library of measured parts and validate its own designs for collisions and clearance.
- Built the **React Three Fiber** viewer that animates each assembly step and deployed it with **Docker Compose** on Vultr.

huddle | *Node.js, Cloudflare Workers, MCP* — *Oct 2026 – Present*

- Designed a coordination layer in **JavaScript** that lets multiple Claude Code agents on separate machines edit one repository at once, using per-file locks that make other checkouts read-only and stream edits to them live.
- Implemented a dependency-free **Node.js** relay server on **Cloudflare Workers** that brokers locks, chat and live file edits between agents.
- Added git history sync between checkouts as chunked bundles, branch-scoped locks, and an end-to-end test suite that runs real Claude sessions against the live server.

Submerge | *GLSL, QML, Python* — *Oct 2026 – Present*

- Developed an animated koi theme for KDE Plasma 5 and 6 in **GLSL** and **QML**, with a shader water simulation and a live system HUD, reaching **~1,290 downloads** on the KDE Store.
- Added power-aware frame pacing for battery life and fixed blank rendering on NVIDIA/Wayland by porting the shaders to GLSL ES.

Awards & Honors

Gold & Silver Medalist, IT Systems Networking, Skills Ontario (Regional & Provincial) — Mar 2026 – May 2026
National Champion x2, 1st in Canada, Cybersecurity, $1,000 scholarship — May 2025, Mar 2026
National Finalist, 2nd in Canada, Cybersecurity, $1,600 scholarship — Mar 2025